Let 'y' be the number of cars of type C_2

Let 'z' be the number of cars of type C_3

$$3x+2y+4z=28$$
$$x+y+2z=13$$
$$2x+2y+z=14$$

Here $$\Delta=\begin{vmatrix}3&2&4\\1&1&2\\2&2&1\end{vmatrix}=-3\neq 0$$

$$\Delta_x=\begin{vmatrix}28&2&4\\13&1&2\\14&2&1\end{vmatrix}=-6$$

$$\Delta_y=\begin{vmatrix}3&28&4\\1&13&2\\2&14&1\end{vmatrix}=-9$$

$$\Delta_z=\begin{vmatrix}3&2&28\\1&1&13\\2&2&14\end{vmatrix}=-12$$

$\therefore$ By Cramer's rule

$$x=\frac{\Delta x}{\Delta}=\frac{-6}{-3}=2$$

$$y=\frac{\Delta y}{\Delta}=\frac{-9}{-3}=3 \qquad z=\frac{\Delta z}{\Delta}=\frac{-12}{-3}=4$$

$\therefore$ The number of cars of each type which can be produced are 2, 3 and 4.

Exercise 1.2

1. Solve the following equations by using Cramer's rule

(i) $2x+3y=7;\quad 3x+5y=9$

(ii) $5x+3y=17;\quad 3x+7y=31$

(iii) $2x+y-z=3,\quad x+y+z=1,$
$x-2y-3z=4$

(iv) $x+y+z=6,\quad 2x+3y-z=5,$
$6x-2y-3z=-7$

(v) $x+4y+3z=2,\quad 2x-6y+6z=-3,$
$5x-2y+3z=-5$

2. A commodity was produced by using 3 units of labour and 2 units of capital, the total cost is ₹ 62. If the commodity had been produced by using 4 units of labour and one unit of capital, the cost is ₹ 56. What is the cost per unit of labour and capital? (Use determinant method).

3. A total of ₹ 8,600 was invested in two accounts. One account earned $4\frac{3}{4}\%$ annual interest and the other earned $6\frac{1}{2}\%$ annual interest. If the total interest for one year was ₹ 431.25, how much was invested in each account? (Use determinant method).

4. At marina two types of games viz., Horse riding and Quad Bikes riding are available on hourly rent. Keren and Benita spent ₹780 and ₹560 during the month of May.

Name	Number of hours		Total amount spent (in ₹)
	Horse Riding	Quad Bike Riding	
Keren	3	4	780
Benita	2	3	560

Find the hourly charges for the two games (rides). (Use determinant method).

5. In a market survey three commodities A, B and C were considered. In finding out the index number some fixed weights were assigned to the three varieties in each of the commodities. The table below provides the information regarding the consumption of three commodities according to the three varieties and also the total weight received by the commodity.

Commodity / Variety	Variety			Total weight
	I	II	III	
A	1	2	3	11
B	2	4	5	21
C	3	5	6	27

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